CURRICULUM VITAE

MINGFENG ZHAO



(+86)18202573587

mingfengzhao@swufe.edu.cn

School of Mathematics
Southwestern University of Finance and Economics
Room B404, Tongbo Building
No. 555 Liutai Avenue, Wenjiang
Chengdu, 611130
P. R. China

RESEARCH INTERESTS

- ❏ Nonlinear Partial Differential Equations
- ❏ Calculus of Variations and Nonlinear Analysis

EDUCATION

- ❏ *Ph.D. in Mathematics*, University of Connecticut, USA, 2009-2014
  - Major Advisor: Changfeng Gui
  - Doctoral Dissertation: *Traveling Wave Solutions to the Allen-Cahn Equations with Fractional Laplacians*
- ❏ *M.S. in Mathematics*, University of Connecticut, USA, 2009-2012.
- ❏ *B.S. in Mathematics*, Hunan University, China, 2005-2009.
  - Course Advisor: Professor Bizhong Yang
  - Thesis Advisor: Professor Yueping Jiang
  - Undergraduate Thesis: *The Algebraic and Geometric Convergences of* $\mathrm{PU}(\mathbb{C}, n, 1)$ *in the Complex Hyperbolic Spaces.*

WORK EXPERIENCE

- ❏ *Associate Professor*, School of Mathematics, Southwestern University of Finance and Economics, China, 2022-Present.

  Instructor of the following undergraduate and graduate mathematics courses:
  - Partial Differential Equations and Methods of Computations, Spring 2026
  - Advanced Linear Algebra, Spring 2024, Spring 2025, Spring 2026
  - Advanced Linear Algebra II, Fall 2023
  - Partial Differential Equations, Spring 2024, Spring 2025, Spring 2026
  - Modern Partial Differential Equations, Fall 2023, Fall 2025

❑ *Assistant Professor*, Center for Applied Mathematics, Tianjin University, China, 2016-2022.

Instructor of the following undergraduate and graduate mathematics courses:

- Math 2754, Partial Differential Equations, Fall 2019
- Math 4591, Premath I, Fall 2022
- Math 5087, Advanced Mathematics 2B, Spring 2019
- Math 4579, Advanced Mathematics 2A, Fall 2018, Fall 2017, Fall 2020, Fall 2021.
- Math 1069, Nonlinear Analysis, Fall 2018.

❑ *Postdoctoral Research and Teaching Fellow*, University of British Columbia, Canada, 2014-2016.

Instructor of the following undergraduate mathematics courses:

- Math 215/255, Ordinary Differential Equations, Fall 2015, Fall 2014.
- Math 257/316, Partial Differential Equations, Summer 2015.
- Math 105, Integral Calculus with Applications to Commerce and Social Sciences, Spring 2015.

❑ *Quantitative Learning Center Tutor*, University of Connecticut, USA, 2010-2011.

Provided tutoring to university students in all undergraduate mathematics courses.

❑ *Graduate Assistant*, University of Connecticut, USA, 2009-2014.

Teaching assistant for the following graduate and undergraduate mathematics courses:

- Math 5210, Abstract Algebra I (Graduate Course), Fall 2013, Fall 2011, Fall 2010.
- Math 5110, Introduction to Modern Analysis (Graduate Course), Fall 2013, Fall 2012, Fall 2010.
- Math 3435, Partial Differential Equations, Fall 2013.
- Math 3151, Analysis II, Spring 2013.
- Math 3150, Analysis I, Spring 2013, Fall 2012.
- Math 3710, Introduction to Mathematical Modeling, Fall 2012.
- Math 3146, Introduction to Complex Variables, Fall 2012.
- Math 2110, Multivariable Calculus, Spring 2012.
- Math 5120, Complex Function Theory I (Graduate Course), Spring 2011.
- Math 5111, Measure and Integration (Graduate Course), Spring 2011.
- Math 2141, Advanced Calculus I, Fall 2011.

PUBLICATIONS

1. **Mingfeng Zhao**, Changfeng Gui and Zhilei Liang, *On conditional Invisicid limit for Navier-Stokes equations in a cubic 3D domain*, submitted, 2025.
2. Huaiqian Li and **Mingfeng Zhao**, *Fractional Dunkl Perimeters: Exact Coarea Formulae, Isoperimetry, and a Strong Critical Besov Embedding*, submitted, 2026.
3. Changfeng Gui and **Mingfeng Zhao**, *Nonglobal solutions and singular asymptotics for a third-order nonlinear ODE*, submitted, 2026.
4. **Mingfeng Zhao**, Changfeng Gui, Hongxia Guo and Ping Lin, *Classification of all solutions to a third order ODE arising from the 2D laminar flow*, Journal of Differential Equations, 484, 2026.
5. Huaiqian Li, and **Mingfeng Zhao**, *Dimension-Free Square Function Estimates for Dunkl Operators*, Mathematische Nachrichten, 296: 1225-243, 2023.

6. Jérôme Coville, Changfeng Gui, and **Mingfeng Zhao**, *Propagation acceleration in reaction diffusion equations with anomalous diffusions*, Nonlinearity, 34(3): 1544-1576, 2021.
7. Hongxia Guo, Changfeng Gui, Ping Lin, and **Mingfeng Zhao**, *Multiple solutions and their asymptotics for laminar flows through a porous channel with different permeabilities*, IMA Journal of Applied Mathematics, 85(2): 280-308, 2020.
8. Nassif Ghoussoub, Frédéric Robert, Shaya Shakerian, and **Mingfeng Zhao**, *Mass and asymptotics associated to fractional Hardy–Schródinger operators in critical regimes*, Communications in Partial Differential Equations, 43(6): 859-892, 2018.
9. Changfeng Gui and **Mingfeng Zhao**, *Asymptotic formula for the speed of traveling wave solutions to Allen-Cahn equations*, Scientia Sinica Mathematica, 46(5): 549-562, 2016.
10. Changfeng Gui and **Mingfeng Zhao**, *Traveling wave solutions to Allen-Cahn Equation with a fractional Laplacian*, Annales de l'Institut Henri Poincare (C) Non Linear Analysis, 32(4): 785-812, 2015.
11. Frank Arthur, Xiaodong Yan, and **Mingfeng Zhao**, *A Liouville-type theorem for higher order elliptic systems*, Discrete and Continuous Dynamical Systems - Series A, 34(9): 3317-3339, 2014.

Research Grants

1. NSFC11801404, Study on Nonlinear Partial Differential Equations with Geometric Structures, January 01, 2026-December 31, 2030
2. 2025ZYD0157, Mathematical Theory of Chemotactic Cross-Diffusion Models in Tumor Evolution, September 01, 2025-August 31, 2028
3. NSFC11801404, Research on the Traveling Wave Solutions involving the Fractional Laplacian, January 01, 2019-December 31, 2021

HONORS & AWARDS

1. *Michael Neumann Dissertation Award*, University of Connecticut, USA, 2014.
2. *Graduate School Predoctoral Fellowship*, University of Connecticut, USA, 2014, 2013, 2010.
3. *Graduate School Dissertation Fellowship*, University of Connecticut, USA, 2014.
4. *Graduate School Doctoral Dissertation Fellowship*, University of Connecticut, USA, 2013.
5. *AMS Graduate Travel Award*, Joint Mathematics Meetings, USA, 2012.
6. *Graduate School Doctoral Travel Grant*, University of Connecticut, USA, 2010.
7. *Outstanding Scholars Graduate Fellowship with Summer Fellowship* (3 years), University of Connecticut, USA, 2009-2012.
8. *Excellent Undergraduate Dissertation Award*, Hunan University, China, 2009.
9. *Hugo Shong Scholarship*, Hunan University, China, 2007.

MISCELLANEOUS

- ❏ Nationality: China.
- ❏ Languages: Mandarin (Native), Sichuanese (Dialect), English.
- ❏ Computer Skills: LaTeX, Matlab, Mathematica, Maple, Sage.
- ❏ Membership: American Mathematical Society, Pi Mu Epsilon (National Mathematics Honorary Society).